

Marx's Mathematical Manuscripts 1881

‘On the Concept of the Derived Function’¹

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I

Let the independent variable x increase to x_1 ; then the dependent variable y increases to y_1 .²

here in I) we consider the simplest possible case, where x appears only to the first power.

1) $y = ax$, when x increases to x_1 ,

$$y_1 = ax_1 \text{ and } y_1 - y = a(x_1 - x)$$

Now allow the *differential operation* to occur, that is, we let x_1 take on the value of x . Then

$$x_1 = x; x_1 - x = 0 ,$$

thus

$$a(x_1 - x) = a \cdot 0 = 0 .$$

Furthermore, since y , only becomes y_1 because x increases to x_1 , we have at the same time

$$y_1 = y ; y_1 - y = 0 .$$

Thus

$$y_1 - y = a(x_1 - x)$$

changes to $0 = 0$.

First making the differentiation and then removing it therefore leads literally to *nothing*. The whole difficulty in understanding the differential operation (as in the *negation of the negation* generally) lies precisely in seeing *how* it differs from such a simple procedure and therefore leads to real results.

If we divide both $a(x_1 - x)$ and the left side of the corresponding equation by the factor $x_1 - x$, we then obtain

$$(y_1 - y)/(x_1 - x) = a .$$

Since y is the *dependent variable*, it cannot carry out any independent motion at all, y_1 therefore cannot equal y and $y_1 - y = 0$ without x_1 first having become equal to x .

On the other hand we have seen that x_1 cannot become equal to x in the function $a(x_1 - x)$ without making the latter $= 0$. The factor $x_1 - x$ was thus *necessarily a finite difference*³ when both sides of the equation were divided by it. At the moment of the construction of the ratio

$$(y_1 - y)/(x_1 - x)$$

$x_1 - x$ is therefore always a finite difference. It follows that

$$(y_1 - y)/(x_1 - x)$$

is a *ratio of finite differences*, and correspondingly

$$(y_1 - y)/(x_1 - x) = \Delta y / \Delta x$$

Therefore

$$(y_1 - y)/(x_1 - x) \text{ or } \Delta y/\Delta x = a ,$$

where the constant a represents the *limite value (Grenzwert)* of the ratio of the finite differences of the variables.⁵

Since a is a constant, no change may take place in it; hence none can occur on the *right-hand side* of the equation, which has been reduced to a . Under such circumstances the *differential process* takes place on the left-hand side

$$(y_1 - y)/(x_1 - x) \text{ or } \Delta y/\Delta x ,$$

and this is characteristic of such simple functions as ax .

If in the denominator of this ratio x_1 decreases so that it approaches x , the limit of its decrease is reached as soon as it becomes x . Here the difference becomes $x_1 - x_1 = x - x = 0$ and therefore also $y_1 - y = y - y = 0$. In this manner we obtain

$$0/0 = a .$$

Since in the expression $0/0$ every trace of its origin and its meaning has disappeared, we replace it with dy/dx , where the finite differences $x_1 - x$ or Δx and $y_1 - y$ or Δy appear symbolised as *cancelled* or *vanished* differences, or $\Delta y/\Delta x$ changes to dy/dx .

Thus

$$dy/dx = a .$$

The closely-held belief of some rationalising mathematicians that dy and dx are quantitatively actually only infinitely small, only approaching $0/0$, is a chimera, which will be shown even more palpably under II).

As for the characteristic mentioned above of the case in question, the limit value (*Grenzwert*) of the finite differences is therefore also at the same time the limit value of the differentials.

2) A second example of the same case is

$$y = x$$

$$y_1 = x_1 ; y_1 - y = x_1 - x ;$$

$$(y_1 - y)/(x_1 - x) \text{ or } \Delta y/\Delta x = 1 ; 0/0 \text{ or } dy/dx = 1 .$$

II

When in $y = f(x)$, the function [of] x appears on the right-hand side of the equation in its *developed algebraic expression*,⁶ we call this expression the *original function of x* , its first modification obtained by means of differentiation the *preliminary 'derived' function of x* and its final form obtained by means of the *process of differentiation* the *'derived' function of x* .⁷

$$1) y = ax^3 + bx^2 + cx - e .$$

If x increases to x_1 , then

$$y_1 = ax_1^3 + bx_1^2 + cx_1 - e ,$$

$$y_1 - y = a(x_1^3 - x^3) + b(x_1^2 - x^2) + c(x_1 - x)$$

$$= a(x_1 - x) (x_1^2 + x_1x + x^2)$$

$$+ b(x_1 - x) (x_1 + x) + c(x_1 - x) .$$

Therefore

$$(y_1 - y)/(x_1 - x) \text{ or } \Delta y/\Delta x = a(x_1^2 + x_1x + x^2) + b(x_1 + x) + c .$$

and the the *preliminary 'derivative'* [is]

$$a(x_1^2 + x_1x + x^2) + b(x_1 + x) + c$$

[and it] is here the the *limit value (Grenzwert)* of the *ratios* of the finite differences; that is, however small these differences may become, the value of $\Delta y/\Delta x$ is given by that *'derivative'*. But this is not the same case as that under I) with the limit value of the

ratios of the differentials.^{*}-

When the variable x_1 is decreased in the function

$$a(x_1^2 + x_1x + x^2) + b(x_1 + x) + c$$

until it has reached the limit of its decrease, that is, has become *the same as x*, [then] x_1^2 is changed to x^2 , x_1x to x^2 , and $x_1 + x$ to $2x$, and we obtain the 'derived' function of x :

$$3ax^2 + 2bx + c.$$

It Is here shown in a striking manner:

First: in order to obtain the 'derivative', x_1 must be set $= x$; therefore in the strict *mathematical sense* $x_1 - x = 0$, with no subterfuge about merely approaching infinitely [closely].

Second: Although we set $x_1 = x$ and therefore $x_1 - x = 0$, nonetheless nothing symbolic appears in the 'derivative'.^{*2} The quantity x_1 , although originally obtained from the variation of x , does not disappear; it is only reduced to its minimum limit value $= x$. It remains in the original functions of x as a newly introduced element which, by means of its combinations partly with itself and partly with the x of the original function, finally produces the 'derivative', that is, the *preliminary derivative reduced to its absolute minimum quantity*.

The reduction of x_1 to x within the first (preliminary) 'derived' function changes the left-hand side [from] $\Delta y/\Delta x$ to $0/0$ or dy/dx , thus:

$$0/0 \text{ or } dy/dx = 3ax^2 + 2bx + c,$$

so that the *derivative* appears as the *limit value* of the ratio of the differentials.

The transcendental or symbolic mistake which appears only on the left-hand side has perhaps already lost its terror since it now appears only as the expression of a process which has established its real content on the right-hand side of the equation.

In the 'derivative'

$$3ax^2 + 2bx + c$$

the variable x exists in a completely different condition than in the original function of x (namely, in $ax^3 + bx^2 + cx - e$). It [this derivative] can therefore itself be treated as an original function in turn, and can become the mother of another 'derivative' by the repeated process of differentiation. This can be repeated as long as the variable x has not been finally removed from one of the 'derivatives'; it therefore continues endlessly in functions of x can only be represented by infinite series, which [is] all too often the case.

The symbols d^2y/dx^2 , d^3y/dx^3 , etc., only display the genealogical register of the 'derivatives' with respect to the original given function of x . They are mysterious only so long as one treats them as the *starting point* of the exercise, instead of a merely *the expression of the successively derived functions of x* . For it indeed appears miraculous that a ratio of vanished quantities should pass through a new high degree of disappearance, while there is nothing wonderful in the fact that $3x^2$, for example, can pass through the process of differentiation as well as its mother x^3 . One could just as well begin with $3x^2$ as with the original function of x .

But *nota bene*. The starting point of the process of differentiation actually is $\Delta y/\Delta x$ only in equations as [above] under I), where x appears only to the first power. Then, however, as was shown under I), the result [is]:

$$\Delta y/\Delta x = a = dy/dx .$$

Here therefore as a matter of fact *no new limit value* is found from the process of differentiation which $\Delta y/\Delta x$ passes through; [a result] which remains possible only so long as the preliminary 'derivative' includes the variable x , so long, therefore, as dy/dx remains the symbol of a real process.^{*3}

Of course, it is in no sense an obstacle, that in the differential calculus the symbols dy/dx , d^2y/dx^2 , etc., and their combinations also appear on the right-hand side of the equation. For one knows as well that such purely symbolic equations only indicate the *operations* which are then to be applied to the real function of variables.

$$2) y = axm .$$

As x becomes x_1 , then $y_1 = ax_1^m$ and

$$y_1 - y = a(x_1^m - x^m)$$

$$= a(x_1 - x) (x_1^{m-1} + x_1^{m-2}x + (x_1^{m-3}x^2 + \text{etc.}$$

up to the term $x_1^{m-m}x^{m-1}$).

Therefore

$$(y_1 - y)/(x_1 - x) \text{ or } \Delta y/\Delta x = a(x_1^{m-1} + x_1^{m-2}x + x_1^{m-3}x^2 + \dots \\ + x_1^{m-m}x^{m-1}).$$

We now apply the process of differentiation to this 'preliminary derivative', so that

$$x_1 = x \text{ or } x_1 - x = 0$$

and

$$x_1^{m-1} \text{ is change into } x^{m-1};$$

$$x_1^{m-2}x \text{ into } x^{m-2}x = x^{m-2+1} = x^{m-1};$$

$$x_1^{m-3}x^2 \text{ into } x^{m-3}x^2 = x^{m-3+2} = x^{m-1},$$

and finally,

$$x_1^{m-m}x^{m-1} \text{ into } x^{m-m}x^{m-1} = x^{0+m-1} = x^{m-1}.$$

We thus obtain the function x^{m-1} m times, and the 'derivative' is therefore max^{m-1} .

Due to the equivalence of $x_1 = x$ within the 'preliminary derivative',^{*4} on the left-hand side $\Delta y/\Delta x$ is changed to o/o or dy/dx ; therefore

$$dy/dx = max^{m-1}.$$

All of the operations of the differential calculus could be treated in this manner, which would however be a damned useless mass of details. Nonetheless here is another example; since up to now the difference $x_1 - x$ appeared *only once* in the function of x and therefore disappeared from the right-hand side by means of the formation of

$$(y_1 - y)/(x_1 - x) \text{ or } \Delta y/\Delta x.$$

This [is] not the case in the following:

$$3) y = a^x;$$

Let x become x_1 . Then

$$y_1 = a^{x_1}$$

Therefore

$$y_1 - y = a^{x_1} - a^x = a^x(a^{x_1-x} - 1).$$

[But]

$$a^{x_1-x} = \{1 + (a - 1)\}^{x_1-x},$$

and

$$\{1 + (a - 1)\}^{x_1-x} =$$

$$1 + (x_1 - x)(a - 1) + ((x_1 - x)(x_1 - x - 1)(a - 1)^2)/1 \cdot 2 + \text{etc.} \quad \textcolor{blue}{8}$$

Therefore

$$y_1 - y = a^x(a^{x_1-x} - 1)$$

$$= a^x \cdot \{ (x_1 - x)(a - 1) + ((x_1 - x)(x_1 - x - 1)(a - 1)^2)/1 \cdot 2 + ((x_1 - x)(x_1 - x - 1)(x_1 - x - 2)(a - 1)^3)/1 \cdot 2 \cdot 3 + \text{etc.} \}$$

∴⁹

$$(y_1 - y)/(x_1 - x) \text{ or } \Delta y/\Delta x =$$

$$a^x \cdot \{(a - 1) + ((x_1 - x - 1)(a - 1)^2)/1 \cdot 2 + ((x_1 - x - 1)(x_1 - x - 2)(a - 1)^3)/1 \cdot 2 \cdot 3 + \text{etc.}\}.$$

Now as $x_1 = x$ and thus $x_1 - x = 0$, we obtain for the 'derivative':

$$a^x \cdot \{(a - 1) - (a - 1)^2/2 + (a - 1)^3/3 - \text{etc.}\}.$$

Thus

$$dy/dx = ax \cdot \{(a - 1) - (a - 1)^2/2 + (a - 1)^3/3 - \text{etc.}\}$$

If we designate the sum of the constants in parentheses A, then

$$dy/dx = Aa^x ;$$

but this A = Napierian logarithm of the number^{*5} a , so that:

$$dy/dx, \text{ or, when we replace } y \text{ by its value: } da^x/dx = \log a \cdot a^x ,$$

and

$$da^x = \log a \cdot a^x dx .$$

Supplementary¹⁰

We have considered

1) cases in which the factor $(x_1 - x)$ [occurs] only once in [the expression which leads to] the '*preliminary derivative*' - i.e. [in] the equation of finite differences¹¹ - so that by means of the division of both sides by $x_1 - x$ in the formation of

$$(y_1 - y)/(x_1 - x) \text{ or } \Delta y/\Delta x$$

this same factor is therefore eliminated from the function of x .

2) (in the example $d(a^x)$) cases in which factors of $(x_1 - x)$ remain after the formation of $\Delta y/\Delta x$.¹²

3) Yet to be considered is the case where the factor $x_1 - x$ is *not directly* obtained from the first difference equation ([which leads to] the ‘preliminary derivative’).

$$y = \sqrt{a^2 + x^2},$$

$$y_1 = \sqrt{a^2 + x_1^2},$$

$$y_1 - y = \sqrt{a^2 + x_1^2} - \sqrt{a^2 + x^2};$$

we divide the function of x , the left-hand side as well, therefore, by $x_1 - x$. Then

$$(y_1 - y)/(x_1 - x) \text{ (or } \Delta y/\Delta x) = (\sqrt{a^2 + x_1^2} - \sqrt{a^2 + x^2})/(x_1 - x).$$

In order to rationalise the numerator, [both] numerator and denominator are multiplied by $\sqrt{a^2 + x_1^2} + \sqrt{a^2 + x^2}$, and we obtain:

$$\begin{aligned} \Delta y/\Delta x &= (a^2 + x_1^2 - (a^2 + x^2))/((x_1 - x)(\sqrt{a^2 + x_1^2} + \sqrt{a^2 + x^2})) \\ &= (x_1^2 - x^2)/((x_1 - x)(\sqrt{a^2 + x_1^2} + \sqrt{a^2 + x^2})) \end{aligned}$$

But

$$\begin{aligned} &(x_1^2 - x^2)/((x_1 - x)(\sqrt{a^2 + x_1^2} + \sqrt{a^2 + x^2})) \\ &= ((x_1 - 1)(x_1 + x))/((x_1 - x)(\sqrt{a^2 + x_1^2} + \sqrt{a^2 + x^2})). \end{aligned}$$

So that:

$$\Delta y/\Delta x = (x_1 + x)/(\sqrt{a^2 + x_1^2} + \sqrt{a^2 + x^2})$$

Now when x_1 becomes $= x$, or $x_1 - x = 0$, then

$$dy/dx = 2x/(2 \cdot \sqrt{a^2 + x^2}) = x/\sqrt{a^2 + x^2}$$

So that

$$dy \text{ or } d \sqrt{a^2 + x^2} = x dx / \sqrt{a^2 + x^2} .$$

¹ The manuscript was written in 1881 for Engels. This is the first work in a series of manuscripts conceived by Marx and devoted to a systematic exposition of his ideas on the nature and history of differential calculus. In this work he introduces his concepts of algebraic differentiation and the corresponding algorithm for finding the derivative for certain classes of functions. On the envelope enclosing the manuscript there is the notation in Marx's handwriting: 'For the General'. This was Engels's nickname in Marx's family because of his articles on military questions. Having acquainted himself with the manuscript, Engels answered Marx in a letter on 18 August 1881 (see p.xxvii). The published German text of the manuscript reproduces exhaustively Marx's text. Some of the preparatory material (drafts and supplements) is published on page 473 of Yanovskaya, 1968. Variant readings from the unpublished drafts are provided in footnotes. The manuscript was published for the first time (not in full) in 1933 in Russian translation in the collection Marxism and Science (Marksizm I estestvoznaniye), Moscow, Partizdat, 1933, pp.5-11; and in the journal Under the Banner of Marxism (Pod znamenem marksizma) No.1, 1933, p.15ff. This is the first time it has been published in German.

² In order to avoid confusion with the designation of derivatives, Marx's notation $x', y', \dots$ for the new values of the variable has been replaced here and in all similar cases by $x_1, y_1, \dots$

³ In keeping with the accepted terminology of the source-books which Marx consulted, a finite difference is here understood always to be a non-zero difference.

⁴ Marx distinguishes in each equation two sides (where now we speak of two parts), the left hand and the right hand which do not always play symmetric roles. On the left-hand side of the equation he frequently places two different, equivalent expressions joined by the conjunction 'or'.

⁵ In the mathematical literature which was at Marx's command the term 'limit' (of a function) had no well-defined meaning and was understood most often as the value the function actually reached at the end of an infinite process in which the argument approached its limiting value (see Appendix I, pp.144-145). Marx devoted an entire rough draft to the criticism of these shortcomings in the manuscript, 'On the Ambiguity of the Terms "Limit" and "Limit Value"' (pp.123-126). In the manuscript before us Marx employs the term 'limit' in a special sense: the expression, given by predefinition, for those values of the independent variable at which it becomes undefined. For Marx, the ratios $\Delta y / \Delta x$ (at $\Delta x = 0$ this is transformed to $0/0$) and dy/dx , interpreted as the symbolic expression of the ratio 'of annulled or vanished differences', that is, of $0/0$, are such expressions. With respect to the ratio $\Delta y / \Delta x$, Marx (influenced to a certain degree by the definitions of this concept in Hind and Lacroix; see Appendix I, p.143) took this to be an expression which is identically equal to this ratio when $\Delta x \neq 0$, but which has been predefined by continuity when the ratio is transformed to $0/0$. The 'limit', at that point, consequently, must be the 'preliminary derivative' (concerning which see p.6 and note 7). Exemplifying this, Marx

writes (on p.6), with respect to the ratio $\Delta y/\Delta x$ where $y = ax^3 + bx^2 + cx + d$: 'The "preliminary derivative" $a(x_1^2 + x_1x + x^2) + b(x_1 + x) + c$ appears here as the limit of a ratio of finite differences; that is, no matter how small we allow the differences to become, the value of $\Delta y/\Delta x$ will always be given by this "derivative".' Later (on p.7), Marx mentions that setting x_1 equal to x , that is, setting $\Delta x = 0$, 'reduces this limit value to its absolute minimum quantity', giving its 'final derivative'. Analogously, by 'the limit of the ratio of differentials' Marx in this manuscript means the 'real' ('algebraic' - see note 6) expression which provides the value for this ratio; in other words, the derived function. Marx writes, however, that in the equation $dy/dx = f(x)$, 'neither of the two sides is the limiting value of the other. They approach one another, not in a limit relationship, but rather in a relationship of equivalence,' (see p.126). But here, the concept of 'limit' (and of 'limit value') is used in another sense, close to the one accepted today. Marx uses the term 'absolute minimal expression' (see, for example, p.125) in a sense even closer to the contemporary concept of limit, when he writes in another passage (see p.68) that it is interchangeable with the category of limit, in the sense given it by Lacroix and in which it has had great significance for mathematical analysis (for Lacroix's definition, see Appendix I pp.151-153).

6 By 'algebraic' Marx understands any expression which contains symbols neither of the derivative nor of differentials. Such a use of the term 'algebraic expression' was characteristic of mathematical literature at the beginning of the 19th century. Marx frequently distinguishes between the concepts 'function of (*von*) x ' and 'function in (*in*) x ', that is, the function as a correspondence and the function as an analytical expression (see p.506 Yanovskaya, 1968). In the manuscript before us he does not adhere to this distinction strictly, speaking most of the time of simply 'the function x (*die Funktion x* [rendered 'the function of x ' in English])', perhaps because he always has in mind only functions given by a certain 'algebraic expression'. He provides a correspondence relating the value of the dependent variable y to the value of the independent variable x by means of the equation $y = f(x)$, where y is the dependent variable and $f(x)$ is an analytic expression with respect to the appearance of the variable x in it.

7 The essence of Marx's method of algebraic differentiation consists of his predefinition (for $x_1 = x$) of the ratio of finite differences (having meaning only when $x_1 = x$), $(f(x_1) - f(x))/(x_1 - x)$ (1) by means of continuity. With this goal in mind he writes down the function $\varphi(x_1, x)$, which coincides with (1) for all $x_1 = x$ and which is continuous as $x_1 \rightarrow x$. Marx calls such a function $\varphi(x_1, x)$ the preliminary derived function of the function $f(x)$, while the function $\varphi(x, x)$, which is obtained from $\varphi(x_1, x)$ under the assumption that $x_1 = x$, he calls the derivative of the function $f(x)$. If this function exists (which is a relevant question for the classes of function under consideration), then it coincides with the present-day concept of the derivative, namely: $\lim_{x_1 \rightarrow x} ((f(x_1) - f(x))/(x_1 - x)) = f'(x)$. Already in Marx's time well-known functions existed for which the operation of differentiation was undefined (see p.117 of the present edition [and note 85, p.211]).

* - In a draft of this work (4146, Pl.4), the following appears: 'On the other hand, the process of differentiation (*Differentialprozess*) now takes place in the preliminary "derived" function of x (on the right-hand side), while any movement of the same process on [the] left-hand side is necessarily prohibited.' - Ed.

^{*2} The draft contains the following statement: ‘Finding “the derivative” from the original function of x proceeds in such a manner, that we first take a *finite differentiation (endliche Differentiation)*; this provides a preliminary “derivative” which is the *limit value (Grenzwert)* of $\Delta y/\Delta x$. The process of differentiation (*Differentialprozess*) to which we then proceed, *reduces this limit value* to its absolute minimum quantity (*Minimalgrösse*). The quantity x_1 introduced in the first differentiation does not disappear ...’ - Ed.

^{*3} The draft (Pl.7) includes this sentence: ‘This can only come about, where the preliminary “derived” function includes the variable x , through whose motion, therefore, another truly new value may be formed, so that dy/dx is the symbol of a real process.’ - Ed.

^{*4} On the right-hand side, that is. - Ed.

^{*5} Original 'root'. - Trans.

⁸ Marx reproduces here the formal expansion of the function into a series which is typical of the mathematics books at his command, having left to one side the questions of the series so obtained and the agreement of the value of the function with the limits of the partial sums.

⁹ $\therefore$: a symbol employed in the manuscripts to stand for the word ‘consequently’. Original: ‘root’. - Trans.

¹⁰ The text entitled ‘Supplementary’ comprises the contents of a separate sheet, appended to the manuscript, of independently numbered pages 1 and (on reverse) 2.

¹¹ By equation of finite differences Marx clearly intends an expression of the form $f(x_1) - f(x) = (x_1 - x) \cdot \varphi(x_1, x)$. See note 7

¹² At this point S[amuel] Moore wrote in pencil ‘*Nicht der Fall, diese Factoren sind $x_1 - x - 1, x_1 - x - 2$ etc.*’ (‘Not the case. These factors are $x_1 - x - 1, x_1 - x - 2$, etc.’.) Obviously Marx intends here not the factors $(x_1 - x)$ but rather the expression $x_1 - x$, and meant to say that the transition to zero of the difference $x_1 - x$, having been preserved in the expression for the preliminary derivative, does not deprive the latter of meaning.

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